Protocol Analysis Summary: Pareto-Efficient Long-Gap Event-Intensive OOD Protocols

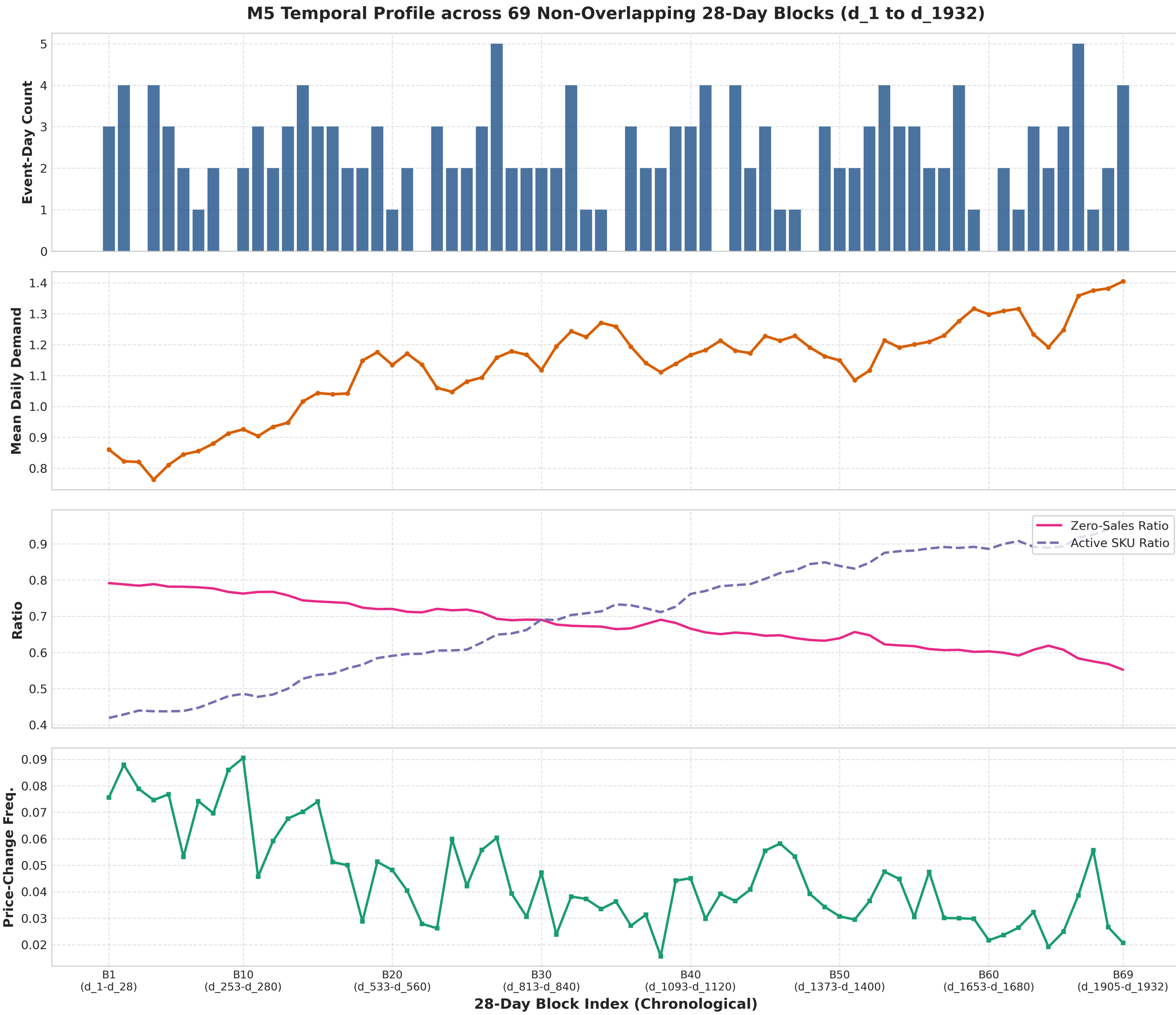
This report presents an extended, model-independent exploratory analysis of the M5 demand forecasting dataset. It identifies **Pareto-efficient candidate OOD target scenarios** that simultaneously achieve:

1. **Genuine temporal separation** from model development; and
2. **High event-day concentration** according to M5 calendar metadata.

No forecasting models were trained, and zero forecasting accuracy metrics (WRMSSE, MAE, MASE) or distribution distance metrics (KS, Wasserstein) were used.

1. M5 Temporal Profile Visualization

The plot below visualizes the chronological progression of key metadata and sales statistics across all 69 non-overlapping 28-day blocks (d_1 to d_{1932}):



2. Pareto Frontier Analysis of Post-ID Candidate OOD Windows

For each model-development cutoff, every eligible rolling 28-day target window occurring post-Validation and post-ID Reference was evaluated. A candidate is **Pareto-efficient** if no other candidate possesses both a longer temporal distance from ID/Val AND a higher or equal event-day count.

Pareto Frontier Summary by Cutoff

A. 3-Year Model Development Cutoff ($T_{train\_end} = 1095$, Jan 27, 2014)

- **Validation**: `d_1096-d_1123` (Feb 2014, 3 event days).
- **ID Reference Target**: `d_1124-d_1151` (Mar 2014, 4 event days).
- **Pareto-Efficient OOD Candidates**:
    i. **`d_1914-d_1941` (Apr 25 � May 22, 2016)**:
        - *Temporal Distance from ID*: **762 days (2.1 years / 25 months)** [Maximal Temporal Gap].

- *Temporal Distance from Val*: **790 days**.
- *Event-Day Count*: **4 event days** (*Cinco De Mayo, Mother's day, OrthodoxEaster, Pesach End*).
- *Event Types*: Cultural: 1, National: 1, Religious: 2.
- *Holiday Context*: Spring holiday cluster.

ii. **`d_1836-d_1863` (Feb 7 � Mar 5, 2016)**:
- *Temporal Distance from ID*: **684 days (1.87 years / 22.5 months)**.
- *Event-Day Count*: **5 event days** (*LentStart, LentWeek2, PresidentsDay, SuperBowl, ValentinesDay*).
- *Event Types*: Cultural: 1, National: 1, Religious: 2, Sporting: 1.

B. End-of-2014 Model Development Cutoff ($T_{train\_end} = 1433$, Dec 31, 2014)

- **Validation**: `d_1434-d_1461` (Jan 2015, 3 event days).
- **ID Reference Target**: `d_1462-d_1489` (Feb 2015, 5 event days).
- **Pareto-Efficient OOD Candidates**:

i. **`d_1914-d_1941` (Apr 25 � May 22, 2016)**:
- *Temporal Distance from ID*: **424 days (1.16 years / 14 months)** [Maximal Temporal Gap].
- *Event-Day Count*: **4 event days** (*Cinco De Mayo, Mother's day, OrthodoxEaster, Pesach End*).

ii. **`d_1836-d_1863` (Feb 7 � Mar 5, 2016)**:
- *Temporal Distance from ID*: **346 days (0.95 years / 11.4 months)**.
- *Event-Day Count*: **5 event days**.

3. Recommended Candidate Protocols for Supervisor Review

We recommend the following **three Pareto-efficient candidate protocols** (each specifying a single, unified Long-Gap Event-Intensive OOD scenario):

Candidate Protocol Comparison Matrix

Protocol Feature	Candidate 1 (Max Distance - Recommended)	Candidate 2 (Balanced Size & Gap)	Candidate 3 (Max Event Density)
Model Cutoff (T_{train_end})	`d_1095` (2014-01-27)	`d_1433` (2014-12-31)	`d_1095` (2014-01-27)
Training History Length	**1,095 days (3.0 years)**	**1,433 days (~3.9 years)**	**1,095 days (3.0 years)**
Validation Window (28d)	`d_1096-d_1123` (3 event days)	`d_1434-d_1461` (3 event days)	`d_1096-d_1123` (3 event days)
ID Reference Target (28d)	`d_1124-d_1151` (4 event days)	`d_1462-d_1489` (5 event days)	`d_1124-d_1151` (4 event days)
Long-Gap Event OOD Target	**`d_1914-d_1941`**	**`d_1914-d_1941`**	**`d_1836-d_1863`**
OOD Calendar Dates	2016-04-25 to 2016-05-22	2016-04-25 to 2016-05-22	2016-02-07 to 2016-03-05

Protocol Feature	Candidate 1 (Max Distance - Recommended)	Candidate 2 (Balanced Size & Gap)	Candidate 3 (Max Event Density)
OOD Event-Day Count	**4 event days**	**4 event days**	**5 event days**
OOD Named Events	Cinco De Mayo, Mother's day, OrthodoxEaster, Pesach End	Cinco De Mayo, Mother's day, OrthodoxEaster, Pesach End	LentStart, LentWeek2, PresidentsDay, SuperBowl, ValentinesDay
Temporal Gap from ID End	**762 days (2.1 yrs / 25 mos)**	**424 days (1.16 yrs / 14 mos)**	**684 days (1.87 yrs / 22.5 mos)**
Temporal Gap from Val End	**790 days**	**452 days**	**712 days**
Pareto Rationale	Maximal temporal distance	Best training size & 1+ yr gap	Highest post-ID event density
Same Checkpoint Evaluation?	**Yes** (Frozen T_{train_end} weights)	**Yes** (Frozen T_{train_end} weights)	**Yes** (Frozen T_{train_end} weights)

4. Evaluation Protocol Design Comparison: Single OOD vs. Dual OOD

We compare the simplified 4-stage protocol:

$$extTraining \longrightarrow extValidation(28d) \longrightarrow extIDReference(28d) \longrightarrow extLong-GapEvent-IntensiveOOD(28d)$$

against the previous 5-stage protocol containing two separate OOD scenarios:

Comparative Analysis Dimensions

1. **Simplicity**:
   - The **simplified single-OOD protocol** requires evaluating only one out-of-domain target scenario (`d_1914-d_1941`).
   - It eliminates redundant evaluation tables, reduces benchmark complexity, and avoids artificial distinctions between "Extended-Gap OOD" vs "Event-Intensive OOD".
2. **Academic Defensibility**:
   - Under earlier cutoffs ($T_{train\_end} = 1095$ or 1433), `d_1914-d_1941` is **simultaneously** genuinely long-gap (762d or 424d temporal separation) AND event-intensive (4 event days).
   - This single window provides a unified, uncompromised test of both temporal degradation and event shock resilience without requiring arbitrary dual-window split rules.
3. **Alignment with Core Research Questions**:
   - The primary research objective is evaluating lightweight student forecasting models and Knowledge Distillation (KD) under distribution shift.
   - Testing on a single robust OOD target directly answers whether Student+KD models outperform standalone Student/Teacher models under combined temporal drift and event stress.
4. **Focus on Lightweight Forecasting Contribution**:
   - Having two separate OOD target scenarios shifts thesis focus heavily onto complex benchmark taxonomy.

- The simplified single-OOD design keeps the experimental structure lean, elegant, and focused on the core Knowledge Distillation contribution.

5. Protocol Invariants & Checkpoint Verification

All three recommended candidate protocols satisfy every required invariant:

- **Chronological Priority**: All Validation, ID, and OOD targets occur strictly post-training ($T_{eval} > T_{train\_end}$).
- **Non-Overlapping Targets**: All evaluation target windows are strictly non-overlapping.
- **90-Day Lookback Availability**: Every forecast origin T_{origin} has a complete 90-day lookback ($T_{origin} - 89$ to T_{origin}) in observed sales data.
- **Frozen Model Checkpoint Evaluation**: Models (TFT Teacher, Student, Student+KD) are trained ONCE on $d_1..T_{train\_end}$. The frozen model checkpoint evaluates both the ID reference target and the Long-Gap Event OOD target by taking the actual preceding 90-day observed sales ($T_{origin} - 89..T_{origin}$) as local encoder context without updating model parameters.
- **Model-Independent Selection**: Scenario dates were selected strictly using calendar metadata and chronology without using model error metrics or predictions.